MATH FOUNDATIONS

Crypto Math Visual Guide

Modular arithmetic, RSA, Diffie-Hellman, and ECC — the math that keeps secrets secret.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Modular Arithmetic: Clock Math

a mod n = remainder after dividing a by n. Wraps around like a clock.

17 mod 12 = 5

like 17:00 = 5 PM

(a+b) mod n

= ((a mod n)+(b mod n)) mod n

Addition, subtraction, multiplication all survive the wrap. Division needs inverses.

MODULAR INVERSE

a^(-1) mod n exists iff gcd(a,n)=1 — division's secret identity

Every public-key system lives inside a modular ring. The clock never lies.

STEP 2/6

Primes and Trapdoor Functions

Easy one way, hard the other — unless you hold the trapdoor.

EASY

multiply two 1024-bit primes -> 2048-bit n

HARD

factor n back into p and q — infeasible

EULER'S THEOREM

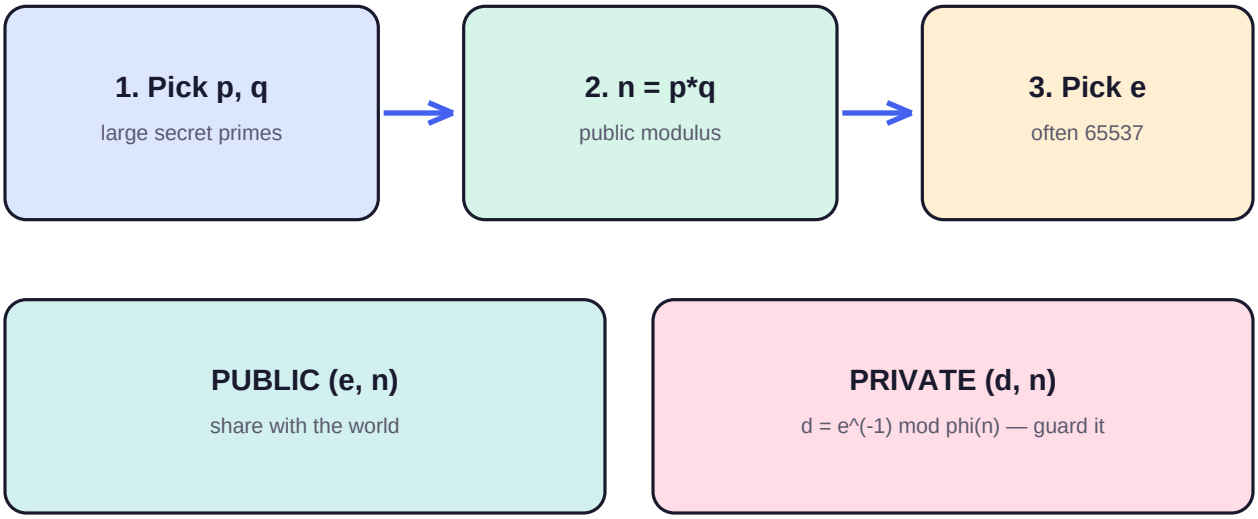
a^phi(n) = 1 mod n — the cycle that makes RSA decryption work

phi(n) = (p-1)(q-1) is easy if you know p, q — impossible-ish if you only know n.

STEP 3/6

RSA Key Generation

Three numbers become a lock anyone can use and only you can open.

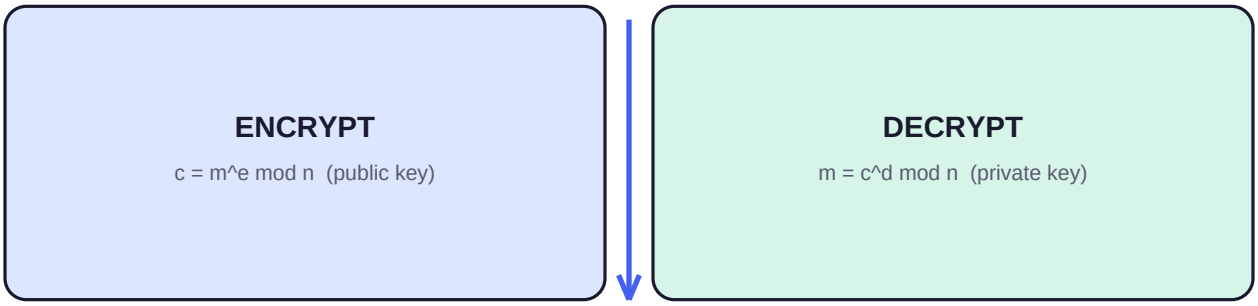


Security = factoring is hard. Quantum computers (Shor's) would break this — hence post-quantum work.

STEP 4/6

RSA Encrypt / Decrypt

Exponentiation modulo n, in both directions.



WHY IT WORKS

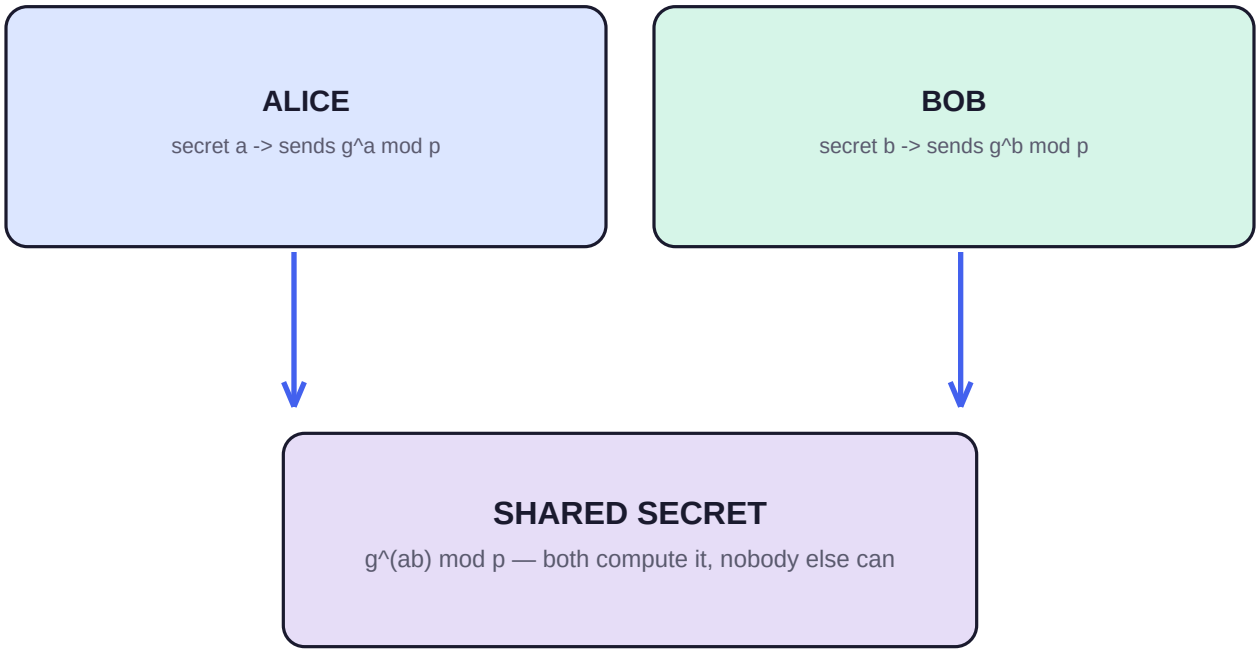
e*d = 1 mod phi(n), so m^(ed) = m mod n by Euler's theorem

In practice RSA encrypts a random session key, not the message — hybrid encryption.

STEP 5/6

Diffie-Hellman Key Exchange

Two strangers agree on a secret over a public channel. Eavesdroppers learn nothing.



Discrete log is the hard problem here. Ephemeral DH gives forward secrecy in TLS 1.3.

STEP 6/6

ECC Intuition

Same discrete-log game, played on an elliptic curve — far smaller keys.

y^2 = x^3 + ax + b (mod p)

points on the curve form a group under 'chord-and-tangent' addition

PRIVATE

secret scalar k

PUBLIC

point k*G — scalar multiplication

256-bit ECC ~= 3072-bit RSA security. Smaller keys, faster handshakes, happier phones.

STEP ✓I✓

Key Takeaways

Crypto Math — Remember

- Modular arithmetic wraps numbers on a clock; inverses exist when gcd(a,n) = 1.
- Trapdoor functions are easy forward, hard backward without the secret — the basis of public-key crypto.
- RSA: pick p, q; publish (e, n); keep d secret. Encrypt m^e, decrypt c^d mod n.
- Diffie-Hellman lets strangers agree on a secret publicly; discrete logs protect it.
- ECC gives RSA-grade security with far smaller keys via elliptic-curve discrete logs.
- Hybrid encryption: public-key crypto protects a session key; symmetric crypto moves the data.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev